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Chen et al.

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(54) **MECHANISM TO CANCEL LTE CRS
INTERFERENCE FROM NEIGHBORING
CELL IN DSS**

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H04W 24/08 (2009.01)

(52) **U.S. Cl.**
CPC **H04L 1/0002** (2013.01); **H04W 16/14**
(2013.01); **H04W 24/08** (2013.01)

(58) **Field of Classification Search**
CPC H04L 1/0002; H04W 16/14; H04W 24/08
See application file for complete search history.

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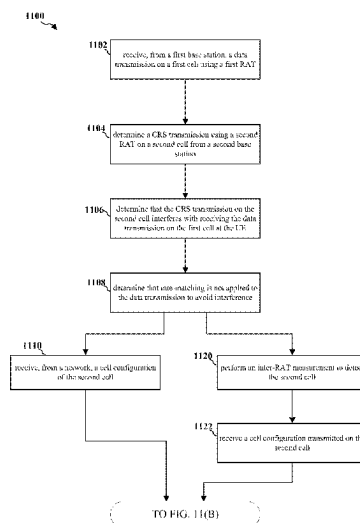
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(57) **ABSTRACT**

In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a UE. The UE receives, from a first base station, a data transmission on a first cell using a first RAT. The UE determines a CRS transmission using a second RAT on a second cell from a second base station. The UE determines that the CRS transmission on the second cell interferes with receiving the data transmission on the first cell at the UE. The UE applies an interference cancellation on the first cell to mitigate an interference from the CRS transmission on the second cell to the data transmission on the first cell.

10 Claims, 15 Drawing Sheets



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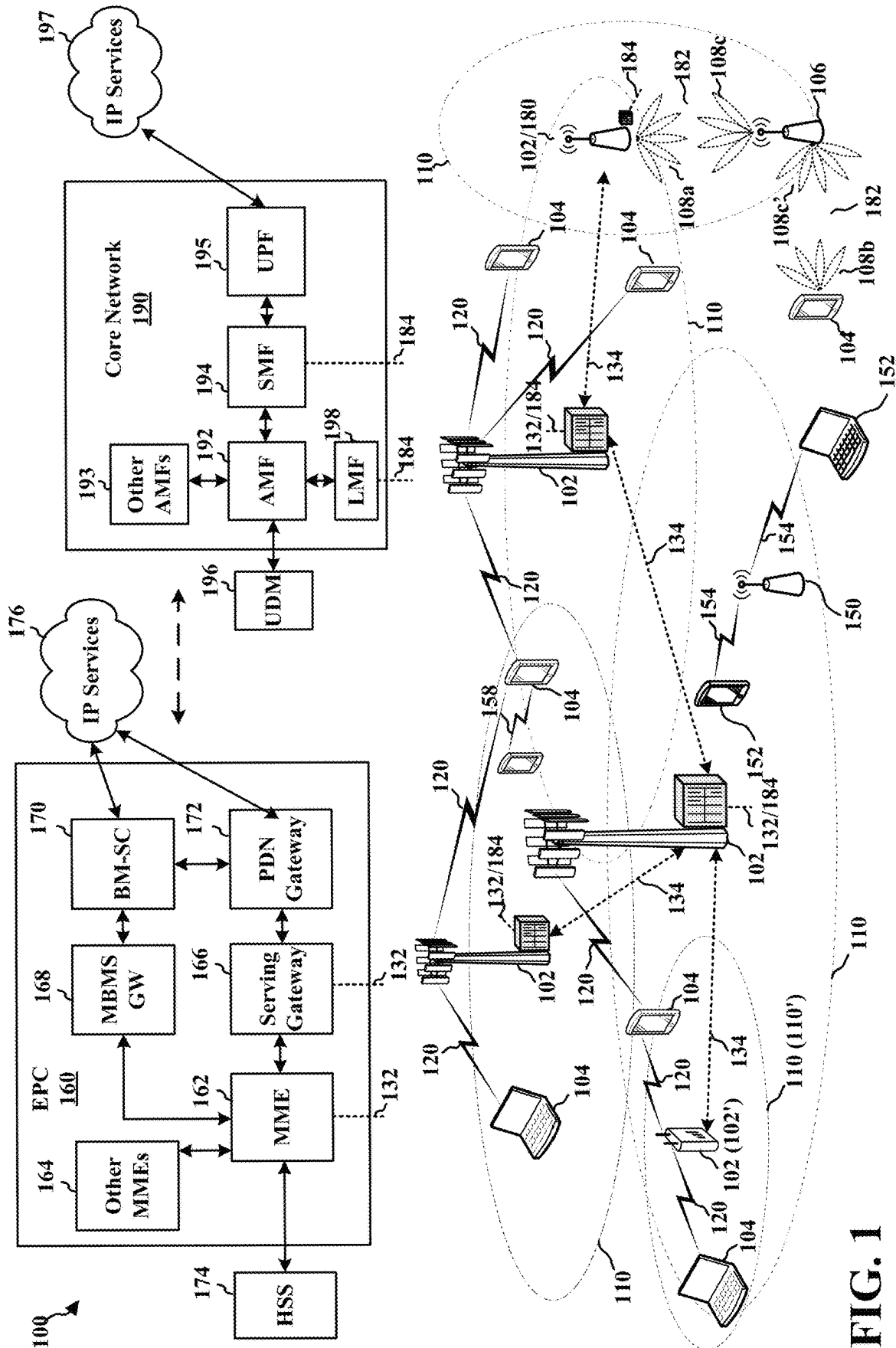


FIG. 1

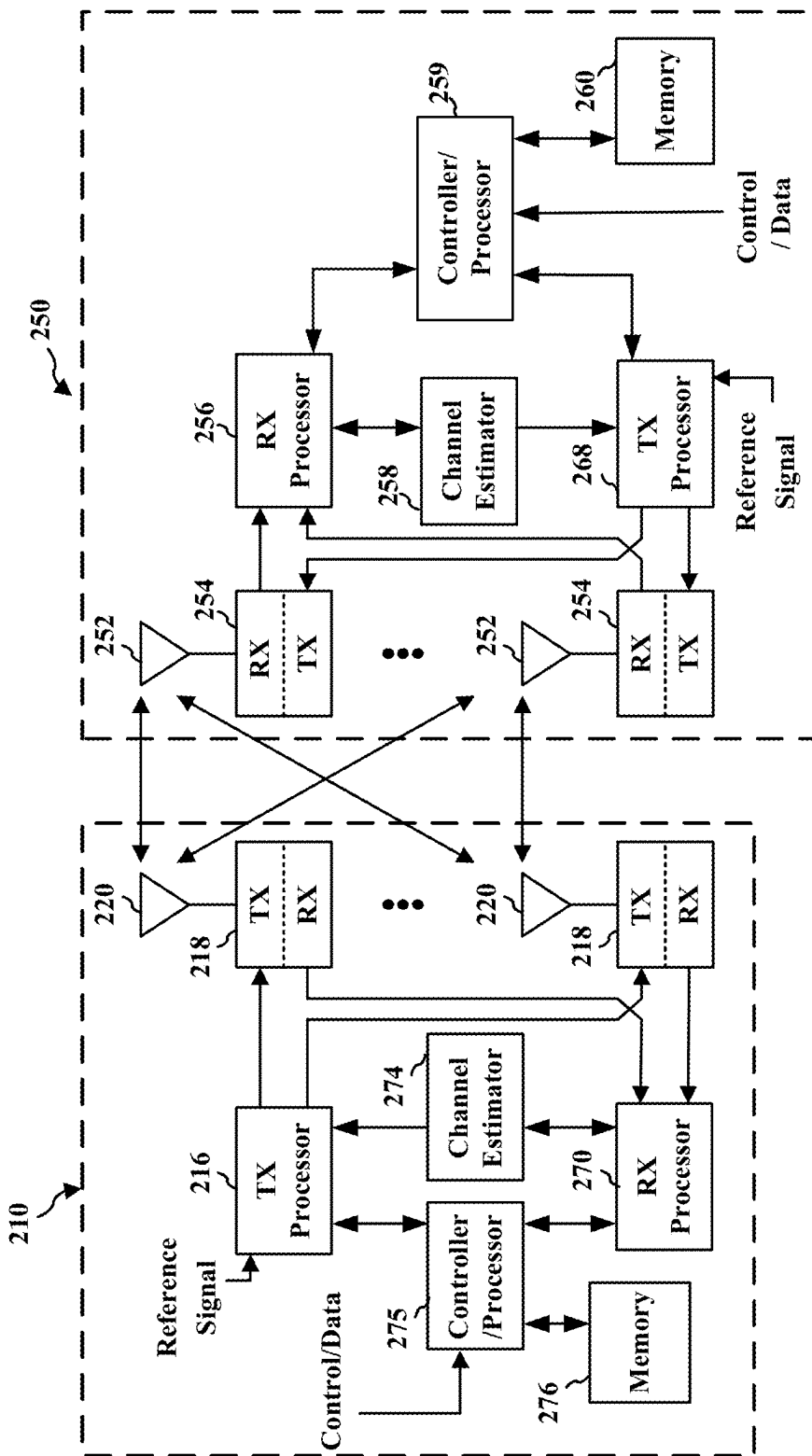


FIG. 2

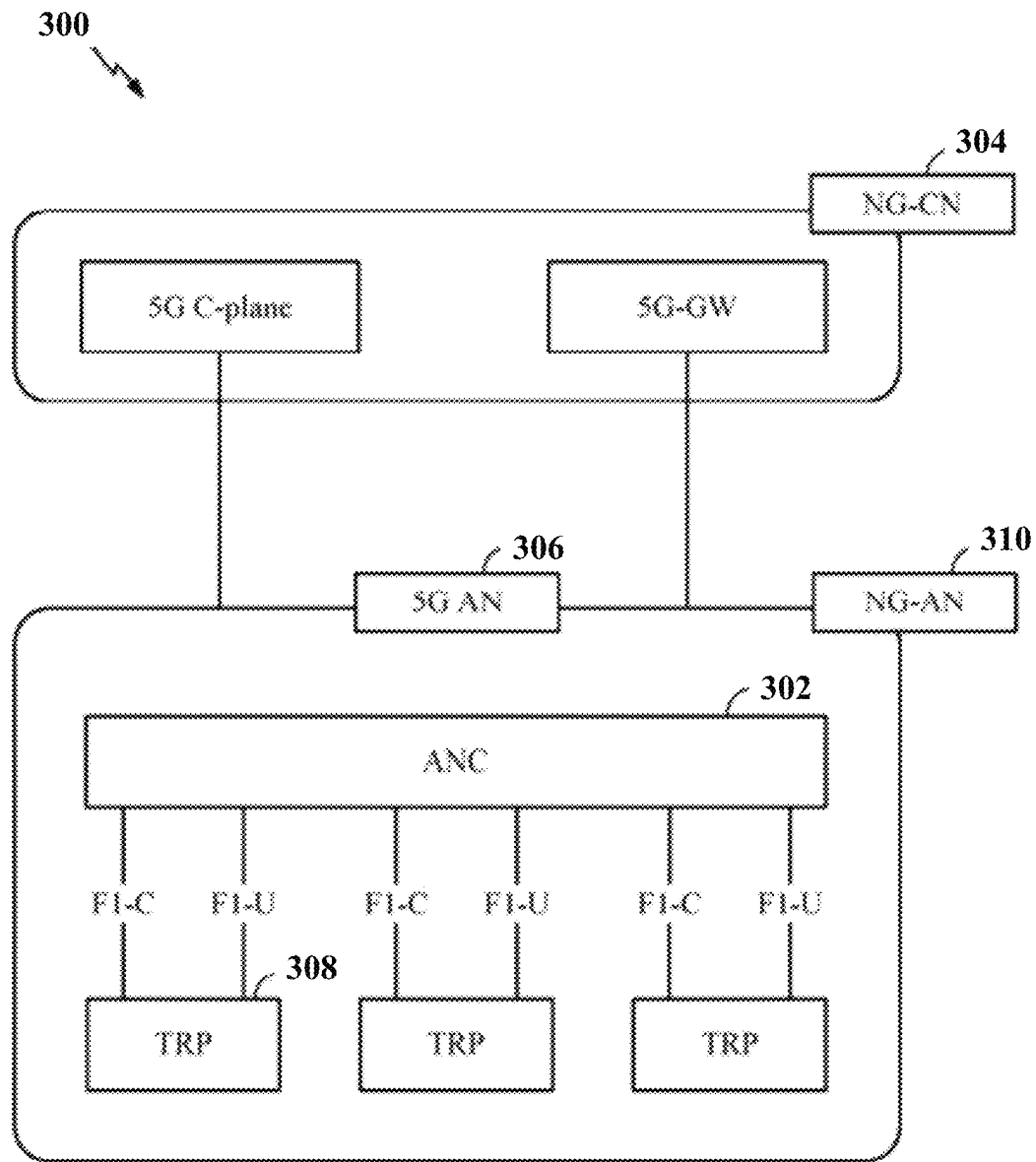


FIG. 3

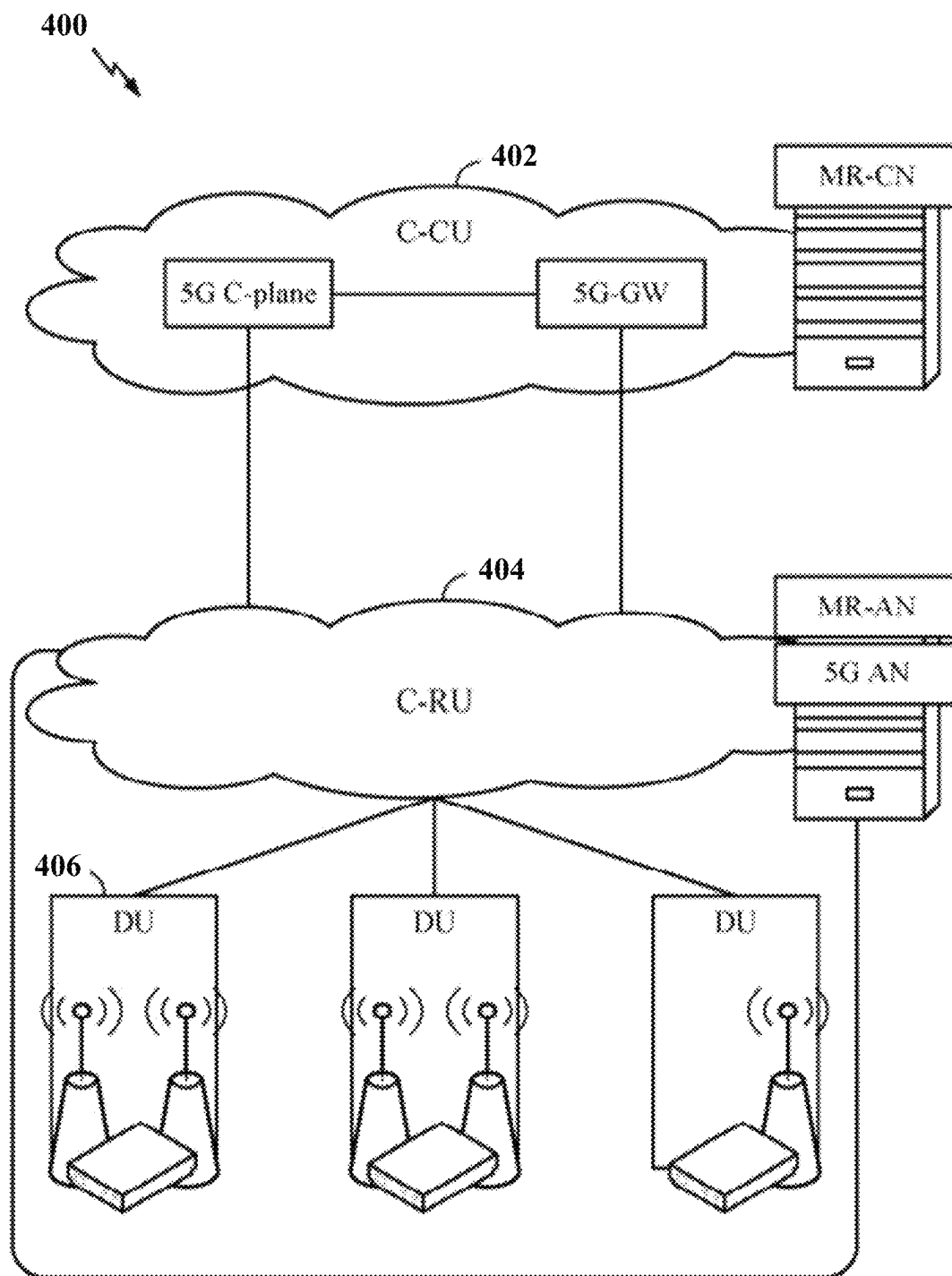


FIG. 4

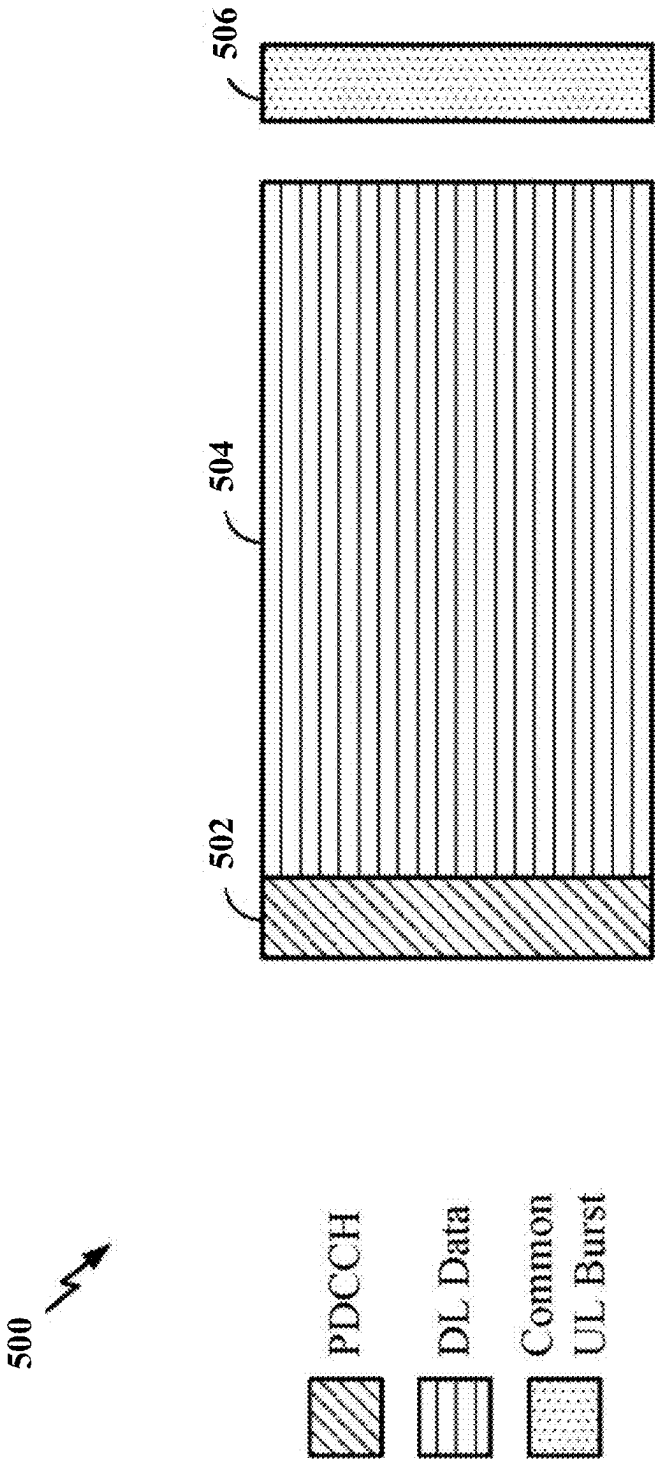


FIG. 5

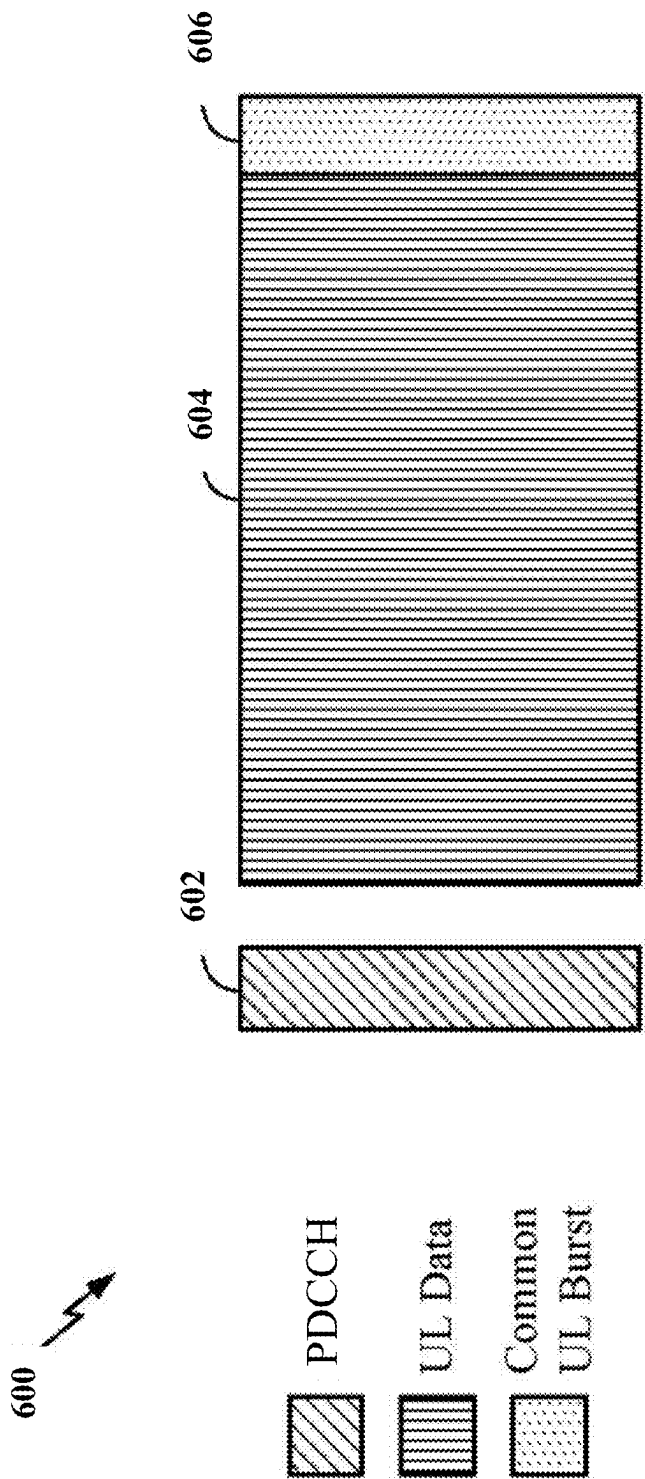


FIG. 6

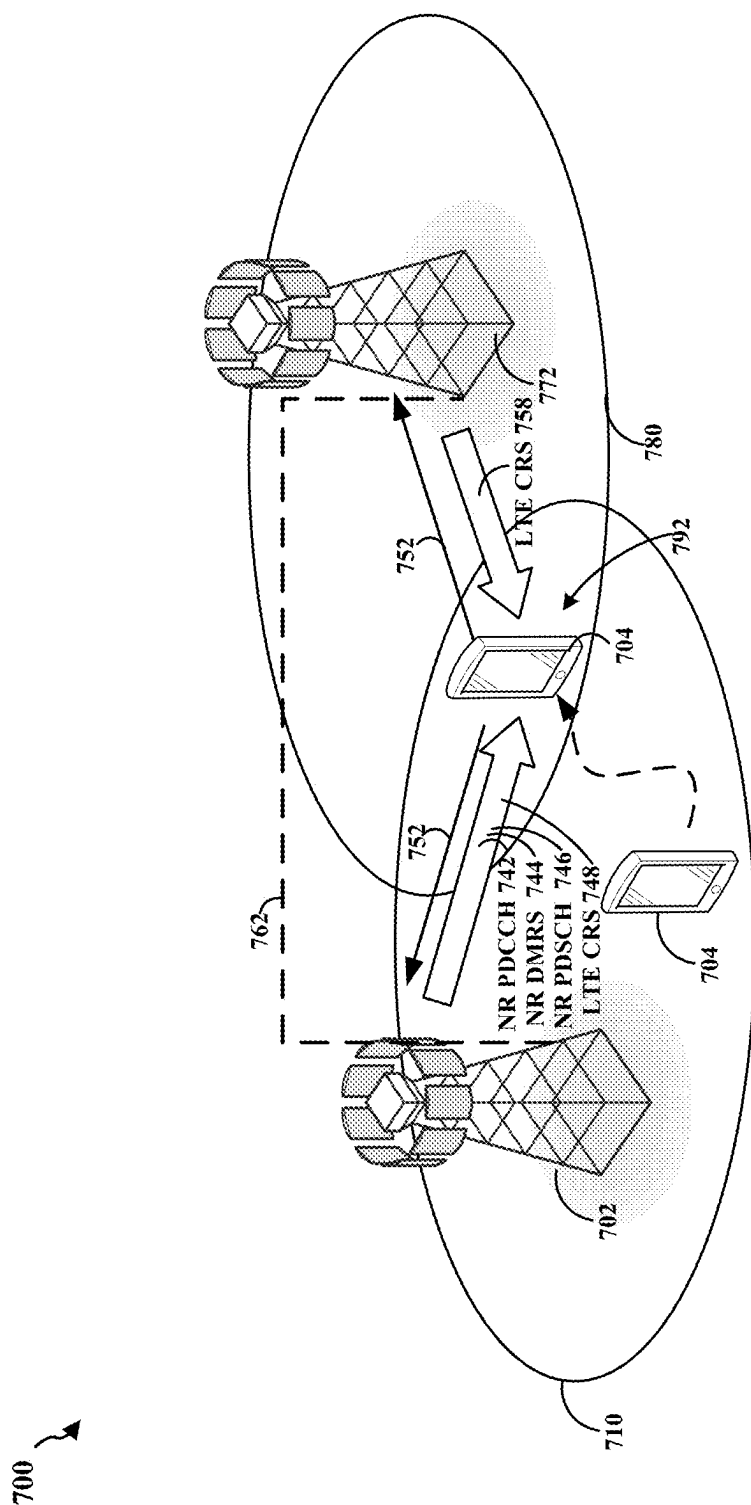


FIG. 7

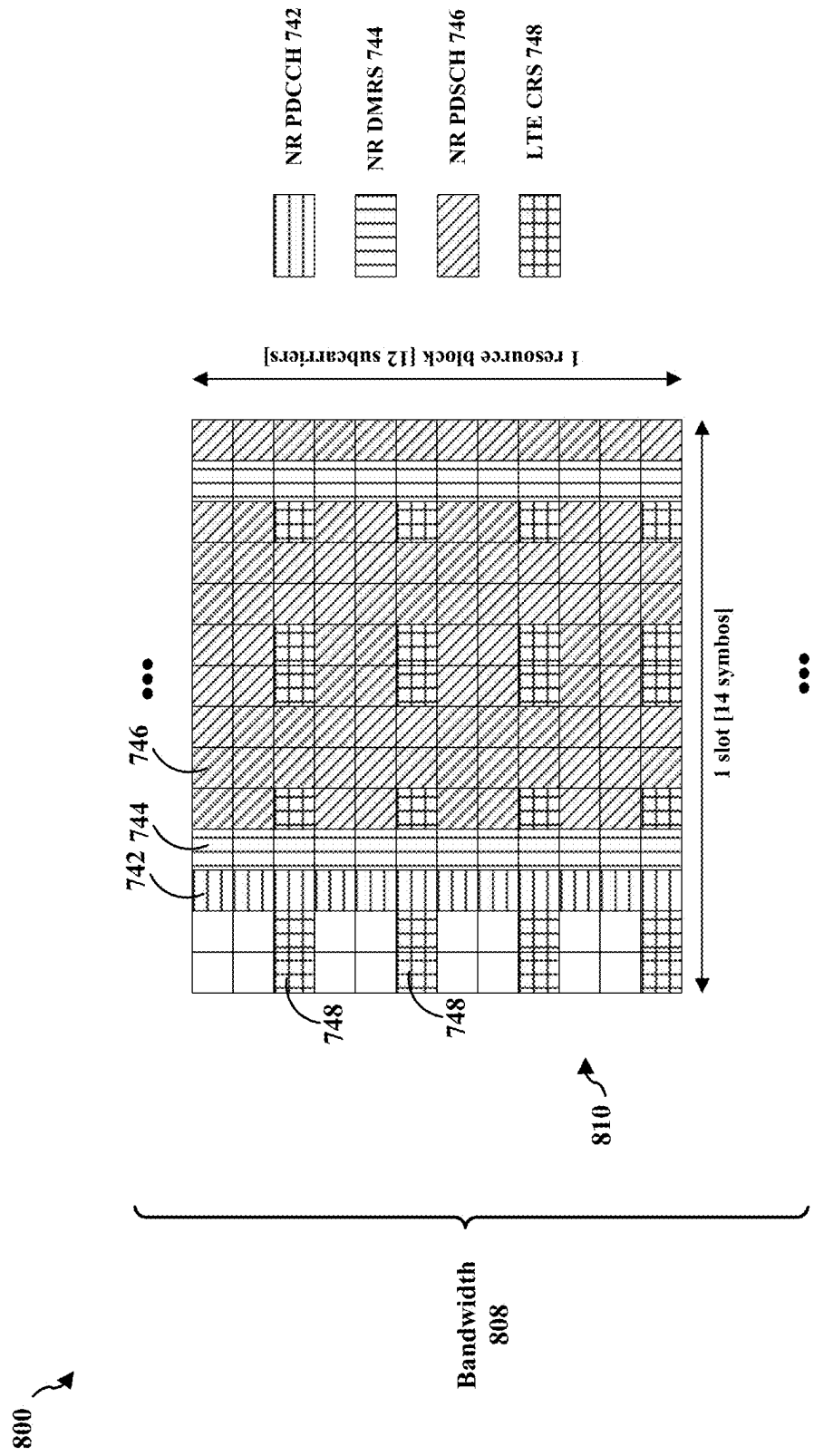


FIG. 8

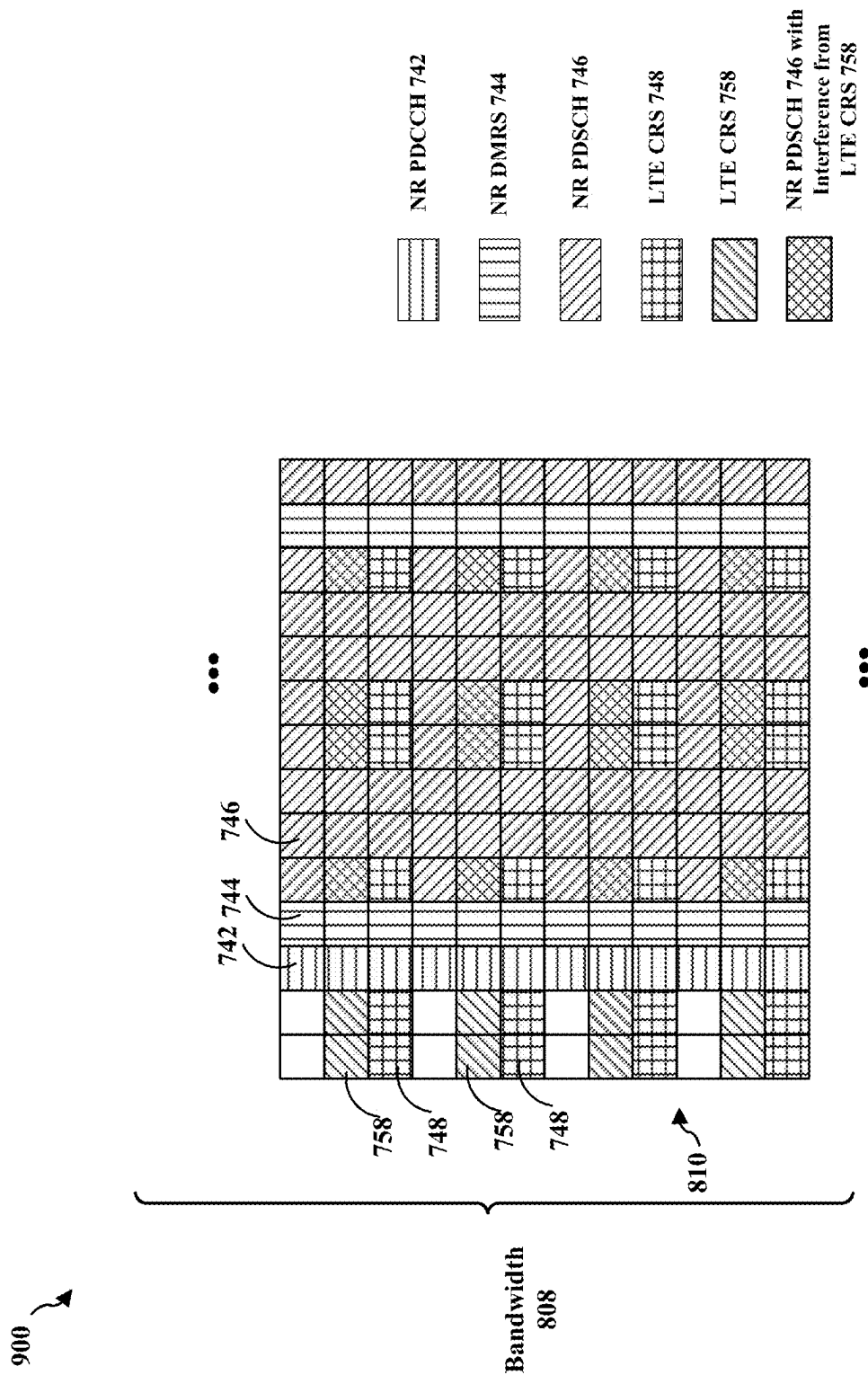


FIG. 9

1000

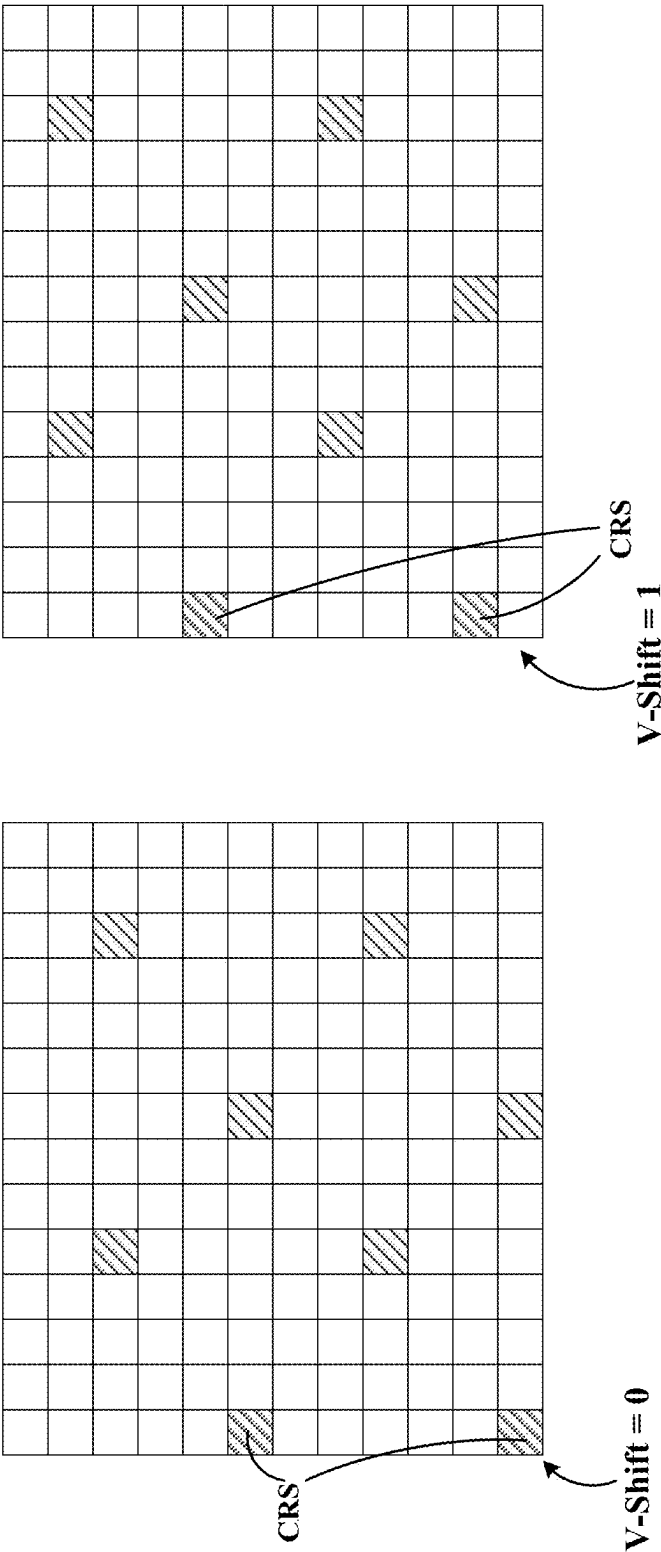
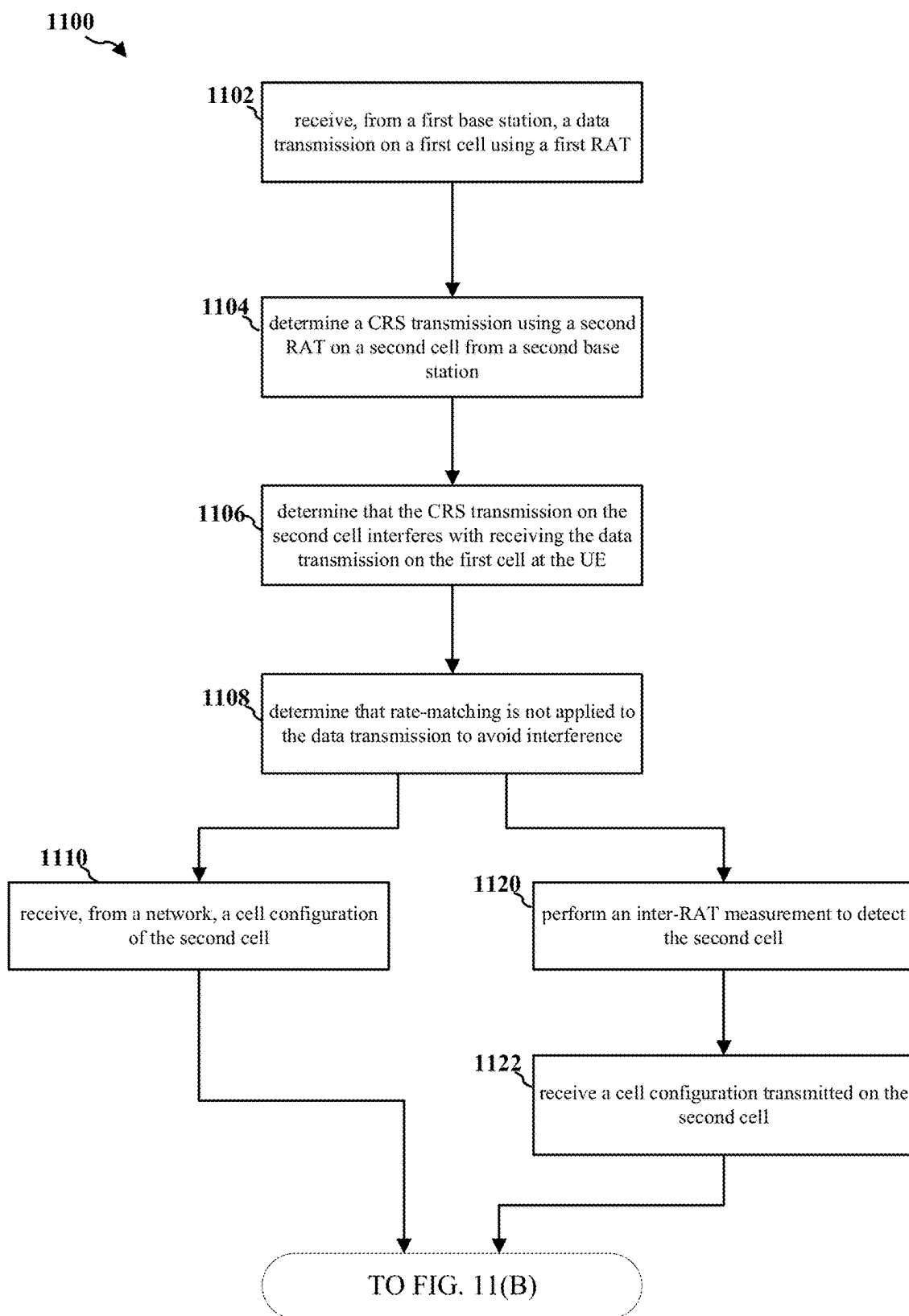
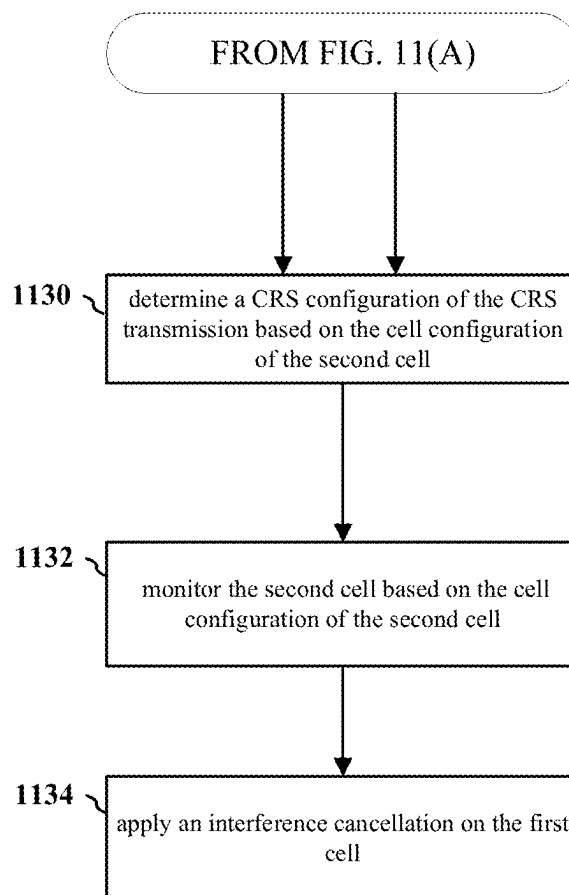
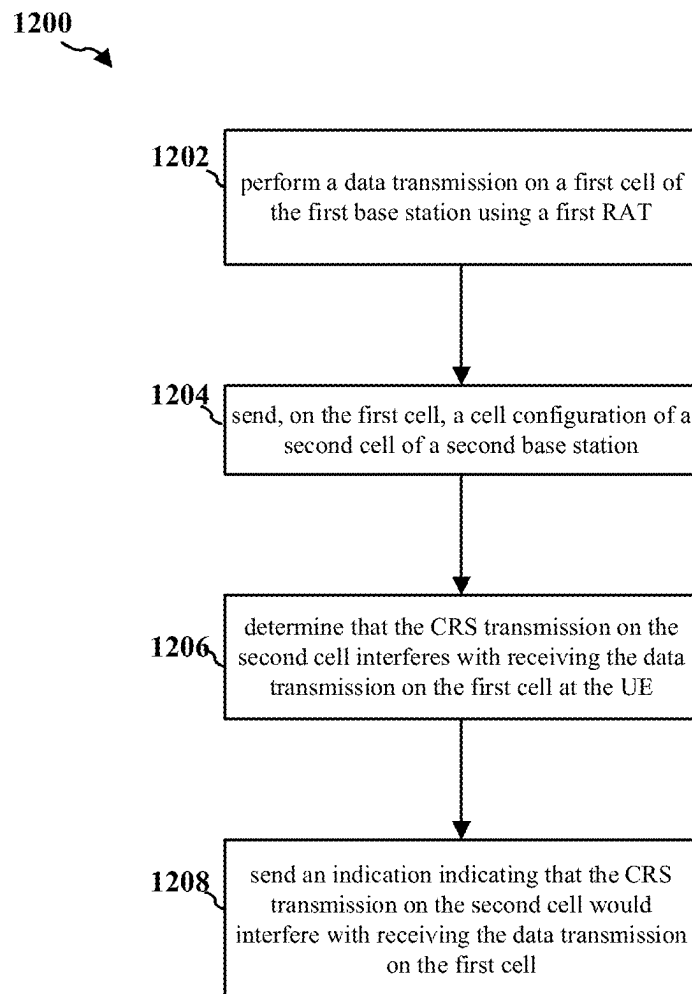


FIG. 10

**FIG. 11(A)**

1100
**FIG. 11(B)**

**FIG. 12**

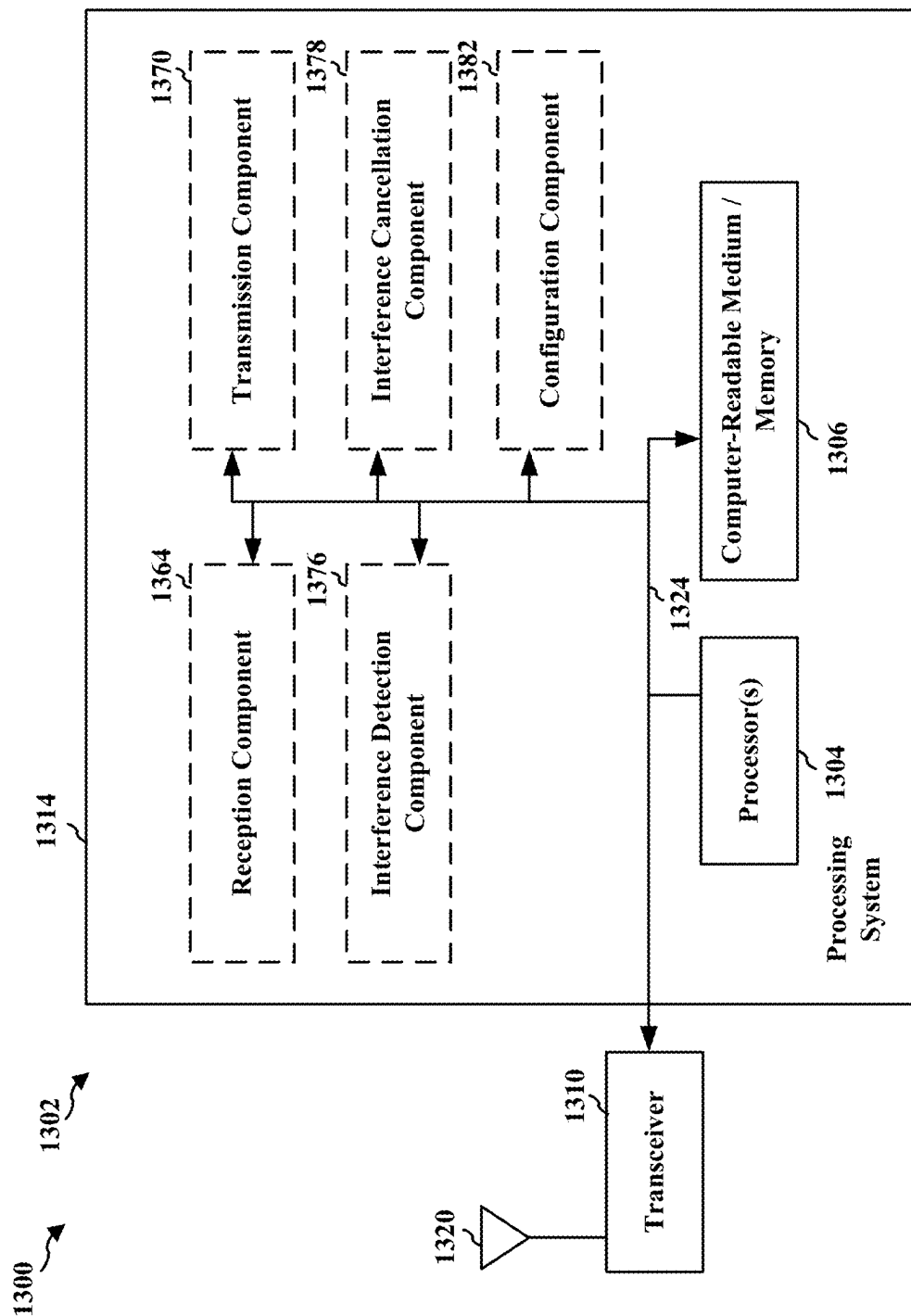


FIG. 13

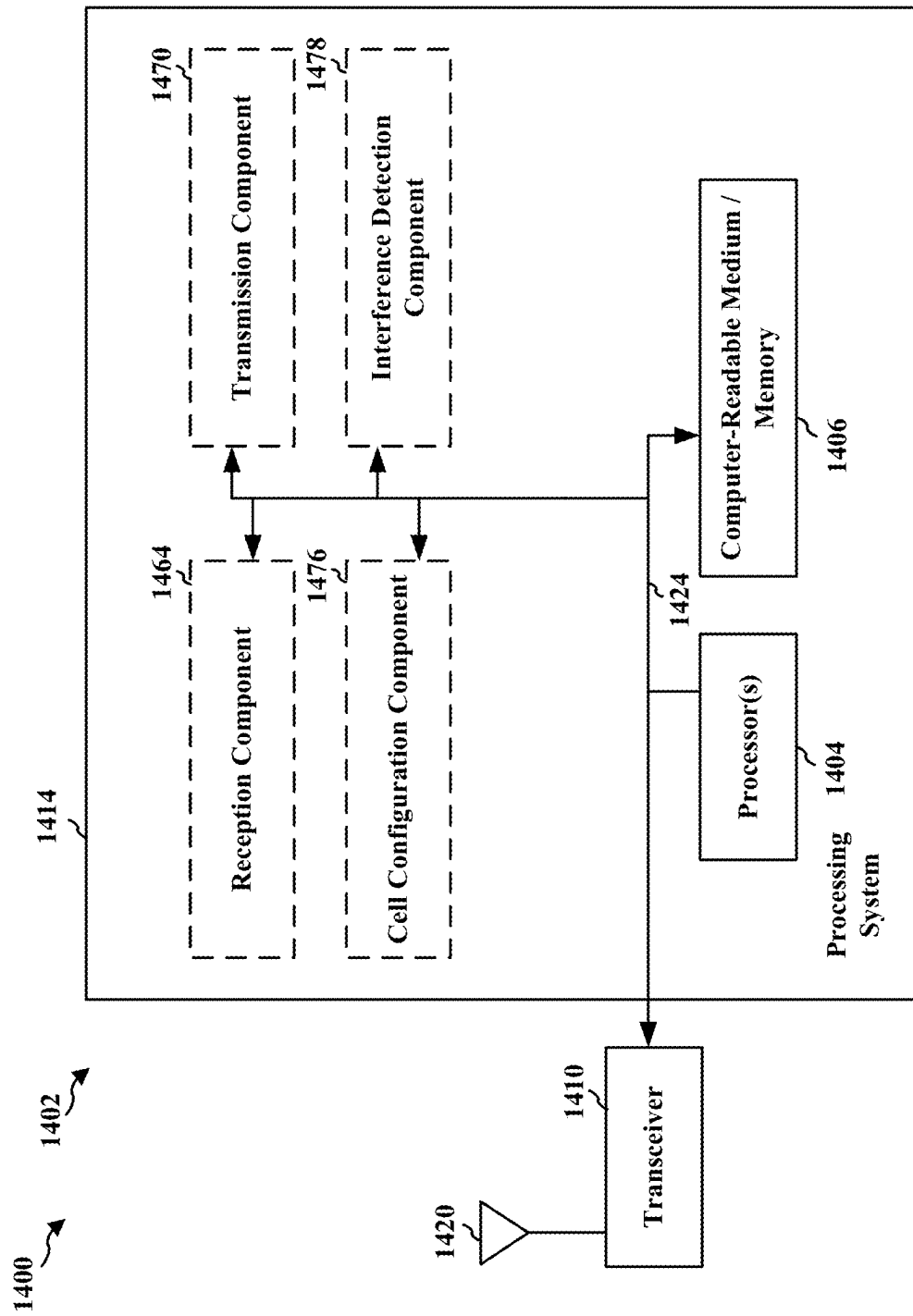


FIG. 14

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MECHANISM TO CANCEL LTE CRS INTERFERENCE FROM NEIGHBORING CELL IN DSS

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims the benefit of U.S. Provisional Application Ser. No. 63/177,469, entitled “MECHANISM TO CANCEL LTE CRS INTERFERENCE FROM NEIGHBORING CELL IN DSS” and filed on Apr. 21, 2021, which is expressly incorporated by reference herein in its entirety.

BACKGROUND

Field

The present disclosure relates generally to communication systems, and more particularly, to techniques of interference mitigation in dynamic spectrum sharing (DSS).

Background

The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

SUMMARY

The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some

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concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a UE. The UE receives, from a first base station, a data transmission on a first cell using a first radio access technology (RAT). The UE determines a cell-specific reference signal (CRS) transmission using a second RAT on a second cell from a second base station. The UE determines that the CRS transmission on the second cell interferes with receiving the data transmission on the first cell at the UE. The UE applies an interference cancellation on the first cell to mitigate an interference from the CRS transmission on the second cell to the data transmission on the first cell.

In another aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may be a first base station. The first base station performs a data transmission on a first cell of the first base station using a first RAT. The second base station sends, on the first cell, a cell configuration of a second cell of a second base station for a UE to detect a CRS transmission on the second cell using a second RAT.

To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating an example of a wireless communications system and an access network.

FIG. 2 is a diagram illustrating a base station in communication with a UE in an access network.

FIG. 3 illustrates an example logical architecture of a distributed access network.

FIG. 4 illustrates an example physical architecture of a distributed access network.

FIG. 5 is a diagram showing an example of a DL-centric slot.

FIG. 6 is a diagram showing an example of an UL-centric slot.

FIG. 7 is a diagram illustrating communications among base stations and UEs.

FIG. 8 is a diagram illustrating a slot of a cell in DSS configuration.

FIG. 9 is a diagram illustrating a slot in an overlapping area of two cells.

FIG. 10 is a diagram illustrating a parameter v -shift used in configuring locations of CRSs.

FIGS. 11 (A) and (B) are a flow chart of a method (process) for mitigating interference.

FIG. 12 is a flow chart of a method (process) for detecting interference.

FIG. 13 is a diagram illustrating an example of a hardware implementation for an apparatus employing a processing system.

FIG. 14 is a diagram illustrating an example of a hardware implementation for another apparatus employing a processing system.

DETAILED DESCRIPTION

The detailed description set forth below in connection with the appended drawings is intended as a description of

various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

Several aspects of telecommunications systems will now be presented with reference to various apparatus and methods. These apparatus and methods will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, etc., whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise.

Accordingly, in one or more example aspects, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the aforementioned types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

FIG. 1 is a diagram illustrating an example of a wireless communications system and an access network 100. The wireless communications system (also referred to as a wireless wide area network (WWAN)) includes base stations 102, UEs 104, an Evolved Packet Core (EPC) 160, and another core network 190 (e.g., a 5G Core (5GC)). The base stations 102 may include macrocells (high power cellular base station) and/or small cells (low power cellular base

station). The macrocells include base stations. The small cells include femtocells, picocells, and microcells.

The base stations 102 configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC 160 through backhaul links 132 (e.g., S1 interface). The base stations 102 configured for 5G NR (collectively referred to as Next Generation RAN (NG-RAN)) may interface with core network 190 through backhaul links 184. In addition to other functions, the base stations 102 may perform one or more of the following functions: transfer of user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, radio access network (RAN) sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations 102 may communicate directly or indirectly (e.g., through the EPC 160 or core network 190) with each other over backhaul links 134 (e.g., X2 interface). The backhaul links 134 may be wired or wireless.

The base stations 102 may wirelessly communicate with the UEs 104. Each of the base stations 102 may provide communication coverage for a respective geographic coverage area 110. There may be overlapping geographic coverage areas 110. For example, the small cell 102' may have a coverage area 110' that overlaps the coverage area 110 of one or more macro base stations 102. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links 120 between the base stations 102 and the UEs 104 may include uplink (UL) (also referred to as reverse link) transmissions from a UE 104 to a base station 102 and/or downlink (DL) (also referred to as forward link) transmissions from a base station 102 to a UE 104. The communication links 120 may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations 102/UEs 104 may use spectrum up to 7 MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

Certain UEs 104 may communicate with each other using device-to-device (D2D) communication link 158. The D2D communication link 158 may use the DL/UL WWAN spectrum. The D2D communication link 158 may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D

communications systems, such as for example, FlashLinQ, WiMedia, Bluetooth, ZigBee, Wi-Fi based on the IEEE 802.11 standard, LTE, or NR.

The wireless communications system may further include a Wi-Fi access point (AP) **150** in communication with Wi-Fi stations (STAs) **152** via communication links **154** in a 5 GHz unlicensed frequency spectrum. When communicating in an unlicensed frequency spectrum, the STAs **152**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

The small cell **102'** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell **102'** may employ NR and use the same 5 GHz unlicensed frequency spectrum as used by the Wi-Fi AP **150**. The small cell **102'**, employing NR in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network.

A base station **102**, whether a small cell **102'** or a large cell (e.g., macro base station), may include an eNB, gNodeB (gNB), or another type of base station. Some base stations, such as gNB **180** may operate in a traditional sub 6 GHz spectrum, in millimeter wave (mmW) frequencies, and/or near mmW frequencies in communication with the UE **104**. When the gNB **180** operates in mmW or near mmW frequencies, the gNB **180** may be referred to as an mmW base station. Extremely high frequency (EHF) is part of the RF in the electromagnetic spectrum. EHF has a range of 30 GHz to 300 GHz and a wavelength between 1 millimeter and 10 millimeters. Radio waves in the band may be referred to as a millimeter wave. Near mmW may extend down to a frequency of 3 GHz with a wavelength of 100 millimeters. The super high frequency (SHF) band extends between 3 GHz and 30 GHz, also referred to as centimeter wave. Communications using the mmW/near mmW radio frequency band (e.g., 3 GHz-300 GHz) has extremely high path loss and a short range. The mmW base station **180** may utilize beamforming **182** with the UE **104** to compensate for the extremely high path loss and short range.

The base station **180** may transmit a beamformed signal to the UE **104** in one or more transmit directions **108a**. The UE **104** may receive the beamformed signal from the base station **180** in one or more receive directions **108b**. The UE **104** may also transmit a beamformed signal to the base station **180** in one or more transmit directions. The base station **180** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **180**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **180**/UE **104**. The transmit and receive directions for the base station **180** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.

The EPC **160** may include a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and a Packet Data Network (PDN) Gateway **172**. The MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. The MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, the MME **162** provides bearer and connection management. All user Internet protocol (IP) packets are transferred through the Serving Gateway **166**, which itself is connected to the PDN Gateway **172**. The PDN Gateway **172** provides UE IP address allocation as well as other functions. The PDN Gateway **172** and the BM-SC **170** are connected

to the IP Services **176**. The IP Services **176** may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services. The BM-SC **170** may provide functions for MBMS user service provisioning and delivery. The BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and may be used to schedule MBMS transmissions. The MBMS Gateway **168** may be used to distribute MBMS traffic to the base stations **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

The core network **190** may include a Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a location management function (LMF) **198**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. The AMF **192** may be in communication with a Unified Data Management (UDM) **196**. The AMF **192** is the control node that processes the signaling between the UEs **104** and the core network **190**. Generally, the SMF **194** provides QoS flow and session management. All user Internet protocol (IP) packets are transferred through the UPF **195**. The UPF **195** provides UE IP address allocation as well as other functions. The UPF **195** is connected to the IP Services **197**. The IP Services **197** may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services.

The base station may also be referred to as a gNB, Node B, evolved Node B (eNB), an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), or some other suitable terminology. The base station **102** provides an access point to the EPC **160** or core network **190** for a UE **104**. Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology.

Although the present disclosure may reference 5G New Radio (NR), the present disclosure may be applicable to other similar areas, such as LTE, LTE-Advanced (LTE-A), Code Division Multiple Access (CDMA), Global System for Mobile communications (GSM), or other wireless/radio access technologies.

FIG. 2 is a block diagram of a base station **210** in communication with a UE **250** in an access network. In the DL, IP packets from the EPC **160** may be provided to a controller/processor **275**. The controller/processor **275** implements layer 3 and layer 2 functionality. Layer 3 includes a radio resource control (RRC) layer, and layer 2

includes a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor 275 provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

The transmit (TX) processor 216 and the receive (RX) processor 270 implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor 216 handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator 274 may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE 250. Each spatial stream may then be provided to a different antenna 220 via a separate transmitter 218TX. Each transmitter 218TX may modulate an RF carrier with a respective spatial stream for transmission.

At the UE 250, each receiver 254RX receives a signal through its respective antenna 252. Each receiver 254RX recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor 256. The TX processor 268 and the RX processor 256 implement layer 1 functionality associated with various signal processing functions. The RX processor 256 may perform spatial processing on the information to recover any spatial streams destined for the UE 250. If multiple spatial streams are destined for the UE 250, they may be combined by the RX processor 256 into a single OFDM symbol stream. The RX processor 256 then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each sub-

carrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station 210. These soft decisions may be based on channel estimates computed by the channel estimator 258. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station 210 on the physical channel. The data and control signals are then provided to the controller/processor 259, which implements layer 3 and layer 2 functionality.

The controller/processor 259 can be associated with a memory 260 that stores program codes and data. The memory 260 may be referred to as a computer-readable medium. In the UL, the controller/processor 259 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the EPC 160. The controller/processor 259 is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

Similar to the functionality described in connection with the DL transmission by the base station 210, the controller/processor 259 provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

Channel estimates derived by a channel estimator 258 from a reference signal or feedback transmitted by the base station 210 may be used by the TX processor 268 to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor 268 may be provided to different antenna 252 via separate transmitters 254TX. Each transmitter 254TX may modulate an RF carrier with a respective spatial stream for transmission. The UL transmission is processed at the base station 210 in a manner similar to that described in connection with the receiver function at the UE 250. Each receiver 218RX receives a signal through its respective antenna 220. Each receiver 218RX recovers information modulated onto an RF carrier and provides the information to a RX processor 270.

The controller/processor 275 can be associated with a memory 276 that stores program codes and data. The memory 276 may be referred to as a computer-readable medium. In the UL, the controller/processor 275 provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE 250. IP packets from the controller/processor 275 may be provided to the EPC 160. The controller/processor 275 is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

New radio (NR) may refer to radios configured to operate according to a new air interface (e.g., other than Orthogonal Frequency Divisional Multiple Access (OFDMA)-based air

interfaces) or fixed transport layer (e.g., other than Internet Protocol (IP)). NR may utilize OFDM with a cyclic prefix (CP) on the uplink and downlink and may include support for half-duplex operation using time division duplexing (TDD). NR may include Enhanced Mobile Broadband (eMBB) service targeting wide bandwidth (e.g. 80 MHz beyond), millimeter wave (mmW) targeting high carrier frequency (e.g. 60 GHz), massive MTC (mMTC) targeting non-backward compatible MTC techniques, and/or mission critical targeting ultra-reliable low latency communications (URLLC) service.

A single component carrier bandwidth of 100 MHz may be supported. In one example, NR resource blocks (RBs) may span 12 sub-carriers with a sub-carrier bandwidth of 60 kHz over a 0.25 ms duration or a bandwidth of 30 kHz over a 0.5 ms duration (similarly, 50 MHz BW for 15 kHz SCS over a 1 ms duration). Each radio frame may consist of 10 subframes (10, 20, 40 or 80 NR slots) with a length of 10 ms. Each slot may indicate a link direction (i.e., DL or UL) for data transmission and the link direction for each slot may be dynamically switched. Each slot may include DL/UL data as well as DL/UL control data. UL and DL slots for NR may be as described in more detail below with respect to FIGS. 5 and 6.

The NR RAN may include a central unit (CU) and distributed units (DUs). A NR BS (e.g., gNB, 5G Node B, Node B, transmission reception point (TRP), access point (AP)) may correspond to one or multiple BSs. NR cells can be configured as access cells (ACells) or data only cells (DCells). For example, the RAN (e.g., a central unit or distributed unit) can configure the cells. DCells may be cells used for carrier aggregation or dual connectivity and may not be used for initial access, cell selection/reselection, or handover. In some cases DCells may not transmit synchronization signals (SS) in some cases DCells may transmit SS. NR BSs may transmit downlink signals to UEs indicating the cell type. Based on the cell type indication, the UE may communicate with the NR BS. For example, the UE may determine NR BSs to consider for cell selection, access, handover, and/or measurement based on the indicated cell type.

FIG. 3 illustrates an example logical architecture of a distributed RAN 300, according to aspects of the present disclosure. A 5G access node 306 may include an access node controller (ANC) 302. The ANC may be a central unit (CU) of the distributed RAN. The backhaul interface to the next generation core network (NG-CN) 304 may terminate at the ANC. The backhaul interface to neighboring next generation access nodes (NG-ANs) 310 may terminate at the ANC. The ANC may include one or more TRPs 308 (which may also be referred to as BSs, NR BSs, Node Bs, 5G NBs, APs, or some other term). As described above, a TRP may be used interchangeably with “cell.”

The TRPs 308 may be a distributed unit (DU). The TRPs may be connected to one ANC (ANC 302) or more than one ANC (not illustrated). For example, for RAN sharing, radio as a service (RaaS), and service specific ANC deployments, the TRP may be connected to more than one ANC. A TRP may include one or more antenna ports. The TRPs may be configured to individually (e.g., dynamic selection) or jointly (e.g., joint transmission) serve traffic to a UE.

The local architecture of the distributed RAN 300 may be used to illustrate fronthaul definition. The architecture may be defined that support fronthauling solutions across different deployment types. For example, the architecture may be based on transmit network capabilities (e.g., bandwidth, latency, and/or jitter). The architecture may share features

and/or components with LTE. According to aspects, the next generation AN (NG-AN) 310 may support dual connectivity with NR. The NG-AN may share a common fronthaul for LTE and NR.

The architecture may enable cooperation between and among TRPs 308. For example, cooperation may be preset within a TRP and/or across TRPs via the ANC 302. According to aspects, no inter-TRP interface may be needed/present.

According to aspects, a dynamic configuration of split logical functions may be present within the architecture of the distributed RAN 300. The PDCP, RLC, MAC protocol may be adaptably placed at the ANC or TRP.

FIG. 4 illustrates an example physical architecture of a distributed RAN 400, according to aspects of the present disclosure. A centralized core network unit (C-CU) 402 may host core network functions. The C-CU may be centrally deployed. C-CU functionality may be offloaded (e.g., to advanced wireless services (AWS)), in an effort to handle peak capacity. A centralized RAN unit (C-RU) 404 may host one or more ANC functions. Optionally, the C-RU may host core network functions locally. The C-RU may have distributed deployment. The C-RU may be closer to the network edge. A distributed unit (DU) 406 may host one or more TRPs. The DU may be located at edges of the network with radio frequency (RF) functionality.

FIG. 5 is a diagram 500 showing an example of a DL-centric slot. The DL-centric slot may include a control portion 502. The control portion 502 may exist in the initial or beginning portion of the DL-centric slot. The control portion 502 may include various scheduling information and/or control information corresponding to various portions of the DL-centric slot. In some configurations, the control portion 502 may be a physical DL control channel (PDCCH), as indicated in FIG. 5. The DL-centric slot may also include a DL data portion 504. The DL data portion 504 may sometimes be referred to as the payload of the DL-centric slot. The DL data portion 504 may include the communication resources utilized to communicate DL data from the scheduling entity (e.g., UE or BS) to the subordinate entity (e.g., UE). In some configurations, the DL data portion 504 may be a physical DL shared channel (PDSCH).

The DL-centric slot may also include a common UL portion 506. The common UL portion 506 may sometimes be referred to as an UL burst, a common UL burst, and/or various other suitable terms. The common UL portion 506 may include feedback information corresponding to various other portions of the DL-centric slot. For example, the common UL portion 506 may include feedback information corresponding to the control portion 502. Non-limiting examples of feedback information may include an ACK signal, a NACK signal, a HARQ indicator, and/or various other suitable types of information. The common UL portion 506 may include additional or alternative information, such as information pertaining to random access channel (RACH) procedures, scheduling requests (SRs), and various other suitable types of information.

As illustrated in FIG. 5, the end of the DL data portion 504 may be separated in time from the beginning of the common UL portion 506. This time separation may sometimes be referred to as a gap, a guard period, a guard interval, and/or various other suitable terms. This separation provides time for the switch-over from DL communication (e.g., reception operation by the subordinate entity (e.g., UE)) to UL communication (e.g., transmission by the subordinate entity (e.g., UE)). One of ordinary skill in the art will understand that the foregoing is merely one example of a DL-centric slot

and alternative structures having similar features may exist without necessarily deviating from the aspects described herein.

FIG. 6 is a diagram 600 showing an example of an UL-centric slot. The UL-centric slot may include a control portion 602. The control portion 602 may exist in the initial or beginning portion of the UL-centric slot. The control portion 602 in FIG. 6 may be similar to the control portion 502 described above with reference to FIG. 5. The UL-centric slot may also include an UL data portion 604. The UL data portion 604 may sometimes be referred to as the payload of the UL-centric slot. The UL portion may refer to the communication resources utilized to communicate UL data from the subordinate entity (e.g., UE) to the scheduling entity (e.g., UE or BS). In some configurations, the control portion 602 may be a physical DL control channel (PDCCH).

As illustrated in FIG. 6, the end of the control portion 602 may be separated in time from the beginning of the UL data portion 604. This time separation may sometimes be referred to as a gap, guard period, guard interval, and/or various other suitable terms. This separation provides time for the switch-over from DL communication (e.g., reception operation by the scheduling entity) to UL communication (e.g., transmission by the scheduling entity). The UL-centric slot may also include a common UL portion 606. The common UL portion 606 in FIG. 6 may be similar to the common UL portion 506 described above with reference to FIG. 5. The common UL portion 606 may additionally or alternatively include information pertaining to channel quality indicator (CQI), sounding reference signals (SRSs), and various other suitable types of information. One of ordinary skill in the art will understand that the foregoing is merely one example of an UL-centric slot and alternative structures having similar features may exist without necessarily deviating from the aspects described herein.

In some circumstances, two or more subordinate entities (e.g., UEs) may communicate with each other using sidelink signals. Real-world applications of such sidelink communications may include public safety, proximity services, UE-to-network relaying, vehicle-to-vehicle (V2V) communications, Internet of Everything (IoE) communications, IoT communications, mission-critical mesh, and/or various other suitable applications. Generally, a sidelink signal may refer to a signal communicated from one subordinate entity (e.g., UE1) to another subordinate entity (e.g., UE2) without relaying that communication through the scheduling entity (e.g., UE or BS), even though the scheduling entity may be utilized for scheduling and/or control purposes. In some examples, the sidelink signals may be communicated using a licensed spectrum (unlike wireless local area networks, which typically use an unlicensed spectrum).

FIG. 7 is a diagram 700 illustrating communications among base stations and UEs. In this example, a base station 702 has established a cell 710 with a UE 704. In particular, the cell 710 may be a dynamic spectrum sharing (DSS) cell. The base station 702 and the UE 704 can use one or more radio access technologies (RATs) for communication concurrently. In particular, the base station 702 can share the spectrum of the cell 710 between the 5G NR and 4G LTE. For example, the base station 702 may transmit LTE CRSs 748, a NR PDCCH 742, NR DMRS 744, and a NR PDSCH 746 on the cell 710. Further, a base station 772 is a neighboring base station of the base station 702. The base station 772 may transmit signals (e.g., LTE CRSs 758) on a cell 780.

FIG. 8 is a diagram 800 illustrating a slot 810 of the cell 710 in DSS configuration. As shown, the base station 702 may transmit the LTE CRSs 748, the NR PDCCH 742, the NR DMRS 744, and the NR PDSCH 746 in resource elements of the slot 810. The cell 710 has a bandwidth 808. In certain configurations, the LTE CRSs 748 overlap with the NR PDSCH 746. As such, the LTE CRSs 748 may cause interference to the reception of the NR PDSCH 746 at the UE 704. The UE 704 may apply rate-matching to the NR PDSCH 746 to puncture the data bits of the NR PDSCH 746 to be transmitted in the resource elements that also contain the LTE CRSs 748. As a result, in those resource elements, only LTE CRSs 748 will be transmitted and no NR PDSCH 746 will be transmitted. Therefore, the interference caused by the LTE CRSs 748 to the reception of the NR PDSCH 746 in those resource elements at the UE 704 can be mitigated.

Referring back to FIG. 7, in certain scenarios, the UE 704 may move into an overlapping area 792 in which the cell 710 and the cell 780 overlap. FIG. 9 is a diagram 900 illustrating the slot 810 in the overlapping area 792, where the resource elements of the slot 810 may contain transmissions from both the cell 710 and the cell 780. In this example, in addition to the LTE CRSs 748, the NR PDCCH 742, the NR DMRS 744, and the NR PDSCH 746 on the cell 710 transmitted from the base station 702, the slot 810 may also contain the LTE CRSs 758 on the cell 780 transmitted from the base station 772. In some of the resource elements of the slot 810, the LTE CRSs 758 overlap with the NR PDSCH 746 and, thus, may cause interference at the UE 704 for receiving the NR PDSCH 746.

Referring to FIGS. 7-9, in certain configurations, the UE 704 may implement procedures to mitigate the interference caused by the LTE CRSs 758 transmission on the cell 780 to its reception of the NR PDSCH 746 on the cell 710. In a first procedure, the UE 704 may determine whether the reception of the NR PDSCH 746 on the cell 710 may be interfered by transmissions from another cell. In a technique A(1), the UE 704 may determine whether the cell 710 is a DSS cell. If the cell 710 is a DSS cell, the transmission on the cell 710 is likely to be interfered by transmission from a neighboring cell (e.g., another DSS cell or LTE cell).

In the technique A(1), the UE 704 may check whether the cell 710 is in a lookup table listing DSS cells, for example, based on a cell ID of the cell 710. If the cell 710 is in the lookup table, the cell 710 is a DSS cell.

In the technique A(1), additionally or alternatively, the UE 704 may determine whether a radio resource control (RRC) information element (IE) received from the base station 702 includes a configuration "lte-CRS-ToMatchAround" and, if yes, whether the "lte-CRS-ToMatchAround" indicates LTE E-UTRA Absolute Radio Frequency Channel Number (EARFCN) associated with the cell 710 or any neighboring cells. The configuration "lte-CRS-ToMatchAround" indicates that UE 704 may need to apply rate-matching around certain LTE CRSs (e.g., the LTE CRSs 748). It can be inferred from the presence of the configuration that the cell 710 is a DSS cell.

In a technique A(2), the base station 772 may receive uplink signals 752 transmitted from the UE 704. The base station 772 can detect that the UE 704 is camped on the cell 710 of the base station 702, e.g., based on information carried in the uplink signals 752. The base station 772 also measures the signal power and the angle of arrival of the uplink signals 752 received. Based on those measurements, the base station 772 can estimate the location of the UE 704. The base station 702 can further determine whether its

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transmission of the LTE CRSs 758 may cause interference at the UE 704 at that location. If yes, the base station 772 can send network-assisted information to the base station 702 through a backhaul 762; the network-assisted information indicates that the transmission from the base station 772 may cause interference at the reception of the UE 704.

In a technique A(3), the UE 704 may determine whether it has received a configuration of a rate-matching pattern. For example, as described supra, the UE 704 may receive a rate-matching pattern from the base station 702 to puncture the data bits of the NR PDSCH 746 to be transmitted in the resource elements that also contain the LTE CRSs 748. As a result, the interference caused by the LTE CRSs 748 to the reception of the NR PDSCH 746 in those resource elements at the UE 704 may be mitigated. In certain circumstances, the UE 704 may also detect the cell 780, but does not receive a rate-matching pattern from the base station 702 to mitigate the interference caused by transmission (e.g., the LTE CRSs 758) on the cell 780. Accordingly, the UE 704 may determine that the transmission from the cell 780 may cause interference to its reception of the NR PDSCH 746 on the cell 710.

In a technique A(4), the UE 704 can measure Reference Signal Received Power (RSRP) and/or Signal to Interference and Noise Ratio (SINR) of signals received from the cell 710 and/or the cell 780. Based on the measurements, the UE 704 may determine whether the signals (e.g., the LTE CRSs 758) from the cell 780 may cause interference to its reception at the cell 710.

By implementing one or more of the techniques A(1) to A(4) described supra, the UE 704 may determine that transmission from the cell 780 or other neighboring cells will cause interference to its reception on the cell 710.

In a second procedure, the UE 704 determines the resource elements in the slot 810 that may have interference. (This allows the UE 704 to enable interference cancellation (IC) at those resource elements in subsequent procedures.) In this example, the UE 704 may use implicit or explicit CRS configuration in neighboring cell information received from the base station 702 to make such a determination. In a technique B(1), the UE 704 receives certain configuration information of the cell 780 (e.g., “neighCellConfig”) from the base station 702. Those configurations, although not explicitly listing the CRSs configurations on the cell 780, may specify subframe configuration and other configurations of the cell 780. Based on those configurations, the UE 704 may estimate the locations of the CRSs in the subframe. As such, the UE 704 can determine the resource elements in the slot 810 that may contain the LTE CRSs 758.

In a technique B(2), the UE 704 may receive explicit CRS configurations in the neighboring cell information received. The neighboring cell information may include one or more of a physical cell ID (PCI) (e.g., the ID of the cell 780), a bandwidth (e.g., the bandwidth of the cell 780), CRS v-shift, CRS antenna port numbers (e.g., the antenna ports of the base station 772 used for transmitting the LTE CRSs 758), and configurations of a Multicast Broadcast Single Frequency Network (MBSFN) subframe. Those configurations may be carried in “CRS-AssistanceInfoList-r11,” “CRS-AssistanceInfoList-r13,” “NeighCellsInfo-r12,” and/or other network configurations sent to the UE 704 from the base station 702. FIG. 10 is a diagram 1000 illustrating the parameter v-shift used in configuring the locations of CRSs. Based on those configurations, the UE 704 may estimate the locations of the CRSs on the cell 780. As such, the UE 704 can determine the resource elements in the slot 810 that contain the LTE CRSs 758.

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In a technique B(3), in addition to the explicit CRS configurations listed in the technique B(2), the base station 702 may send additional CRS configurations to the UE 704. In particular, the CRS configuration may also include a carrier frequency and/or a timing offset. In this example, the configuration of the LTE CRSs 758 may induce a center frequency of a bandwidth of the cell 780. Based on those configurations, the UE 704 may estimate the locations of the CRSs on the cell 780. As such, the UE 704 can determine the resource elements in the slot 810 that contain LTE CRSs 758. The timing offset of two cells indicates the observed timing difference of the two cells (e.g., the cell 710 and the cell 780) in terms of frame boundaries, subframe boundaries, or slot boundaries.

In a third procedure, after the UE 704 determines the locations of the interference, the UE 704 may apply an interference cancellation operation at those locations to mitigate the interference. In a technique C(1), the base station 702 send configurations of the intra-frequency neighboring LTE cells to the UE 704. The configurations may include a PCI, a CRS v-shift, a bandwidth, a carrier frequency (e.g., center frequency), a configuration of frame structure and MBSFN subframe, and/or a timing offset of the neighboring cell (e.g., the cell 780).

In this example, the UE 704 searches and monitors LTE aggressor cells while in locations covered by the cell 710. When UE 704 moves into the overlapping area 792, the UE 704 can detect the cell 780, which is an LTE aggressor cell to the UE 704. Further, the UE 704 has received cell configurations of the cell 780 under the technique C(1). As described supra, the UE 704 may execute the second procedure to determine locations of the LTE CRSs 758. As such, the UE 704 may implement a suitable interference cancellation operation based on the cell configurations of the cell 780 to mitigate the interference caused by the LTE CRSs 758.

In a technique C(2), the UE 704 may not receive cell configuration information of the neighboring cells. The base station 702 may configure the UE 704 with inter-RAT measurement objects for intra-frequency LTE cell measurement (e.g., through “measGapConfig” configuration). Whether or not the UE 704 has received inter-RAT measurement objects, in this example, the UE 704 searches and monitors LTE aggressor cells while in locations covered by the cell 710. When UE 704 moves into the overlapping area 792, the UE 704 can detect the cell 780, which is an LTE aggressor cell to the UE 704. As described supra, the UE 704 may execute the second procedure to determine locations of the LTE CRSs 758. As such, the UE 704 may implement a suitable interference cancellation operation to mitigate the interference caused by the LTE CRSs 758.

FIG. 11 is a flow chart 1100 of a method (process) for mitigating interference. The method may be performed by a UE (e.g., the UE 704). At operation 1102, the UE receives, from a first base station (e.g., the base station 702), a data transmission (e.g., the NR PDSCH 746) on a first cell (e.g., the cell 710) using a first RAT (e.g., 5G NR). At operation 1104, the UE determines a CRS transmission (e.g., the LTE CRSs 758) using a second RAT (e.g., 4G LTE) on a second cell (e.g., the cell 780) from a second base station (e.g., the base station 772). At operation 1106, the UE determines that the CRS transmission on the second cell interferes with receiving the data transmission on the first cell at the UE. In certain configurations, to determine that the CRS transmission interferes with receiving the data transmission, the UE determines that the first cell employs DSS of the first RAT and the second RAT.

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In certain configurations, to determine that the first cell employs DSS, the UE finds an identifier of the first cell in a record of DSS cells. In certain configurations, to determine that the first cell employs DSS, the UE receives, from a network, a configuration for applying rate-matching to the data transmission to avoid interference from a CRS transmission on the first cell using the second RAT.

In certain configurations, to determine that the CRS transmission interferes with receiving the data transmission at the UE, the UE receives, from a network, an indication indicating that the CRS transmission on the second cell would interfere with receiving the data transmission on the first cell. In certain configurations, to determine that the CRS transmission interferes with receiving the data transmission at the UE, the UE receives, from a network, a configuration for applying a given rate-matching to the data transmission to avoid interference from a transmission from a given base station using the second RAT. In certain configurations, to determine that the CRS transmission interferes with receiving the data transmission at the UE, the UE measures a strength of the CRS transmission on the second cell.

At operation **1108**, the UE determines that rate-matching is not applied to the data transmission to avoid interference from the CRS transmission on the second cell to the data transmission on the first cell. Subsequently, in certain configurations, the UE enters operation **1110**, in which the UE receives, from a network, a cell configuration of the second cell. The cell configuration of the second cell includes at least one of: a physical cell ID (PCI), a CRS v-shift, a bandwidth, a CRS antenna port number, and a configuration of a MBSFN subframe. In certain configurations, the cell configuration of the second cell further includes at least one of a center frequency of the second cell and a timing offset. Then, the UE enters operation **1130**.

In certain configurations, subsequent to operation **1108**, the UE enters operation **1120**, in which the UE performs, while receiving the data transmission on the first cell, an inter-RAT measurement to detect the second cell. At operation **1122**, the UE receives a cell configuration transmitted on the second cell. Then, the UE enters operation **1130**.

At operation **1130**, the UE determines a CRS configuration of the CRS transmission based on the cell configuration of the second cell. In certain configurations, the determining the CRS transmission is based on the CRS configuration. At operation **1132**, the UE monitors the second cell based on the cell configuration of the second cell. At operation **1134**, the UE applies, based on the cell configuration, an interference cancellation on the first cell to mitigate an interference from the CRS transmission on the second cell to the data transmission on the first cell.

FIG. **12** is a flow chart **1200** of a method (process) for detecting interference. The method may be performed by a first base station (e.g., the base station **702**). At operation **1202**, the first base station performs a data transmission (e.g., the NR PDSCH **746**) on a first cell (e.g., the cell **710**) of the first base station using a first RAT (e.g., 5G NR). At operation **1204**, the first base station sends, on the first cell, a cell configuration of a second cell (e.g., the cell **780**) of a second base station (e.g., the base station **772**) for a UE (e.g., the UE **704**) to detect a CRS transmission (e.g., the LTE CRSs **758**) on the second cell using a second RAT (e.g., 4G LTE). In certain configurations, the cell configuration of the second cell is sent through a neighboring cell configuration regarding the second cell.

In certain configurations, the cell configuration of the second cell includes at least one of: a PCI, a CRS v-shift, a bandwidth, a CRS antenna port number, and a configuration

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of a MBSFN subframe. In certain configurations, the cell configuration of the second cell further includes at least one of a center frequency of the second cell and a timing offset. In certain configurations, the first base station may send, on the first cell, another cell configuration of the second cell for the UE to apply an interference cancellation to mitigate an interference to receiving the data transmission on the first cell from the CRS transmission on the second cell.

At operation **1206**, the first base station determines that the CRS transmission on the second cell interferes with receiving the data transmission on the first cell at the UE. At operation **1208**, the first base station sends, to the UE, an indication indicating that the CRS transmission on the second cell would interfere with receiving the data transmission on the first cell.

In certain configurations, to determine that the CRS transmission interferes with receiving the data transmission at the UE, the first base station receives an uplink signal from the UE. The first base station measures a strength or an angle of arrival of the uplink signal.

FIG. **13** is a diagram **1300** illustrating an example of a hardware implementation for an apparatus **1302** employing a processing system **1314**. The apparatus **1302** may be a UE. The processing system **1314** may be implemented with a bus architecture, represented generally by a bus **1324**. The bus **1324** may include any number of interconnecting buses and bridges depending on the specific application of the processing system **1314** and the overall design constraints. The bus **1324** links together various circuits including one or more processors and/or hardware components, represented by one or more processors **1304**, a reception component **1364**, a transmission component **1370**, an interference detection component **1376**, an interference cancellation component **1378**, a configuration component **1382**, and a computer-readable medium/memory **1306**. The bus **1324** may also link various other circuits such as timing sources, peripherals, voltage regulators, and power management circuits, etc.

The processing system **1314** may be coupled to a transceiver **1310**, which may be one or more of the transceivers **354**. The transceiver **1310** is coupled to one or more antennas **1320**, which may be the communication antennas **352**.

The transceiver **1310** provides a means for communicating with various other apparatus over a transmission medium. The transceiver **1310** receives a signal from the one or more antennas **1320**, extracts information from the received signal, and provides the extracted information to the processing system **1314**, specifically the reception component **1364**. In addition, the transceiver **1310** receives information from the processing system **1314**, specifically the transmission component **1370**, and based on the received information, generates a signal to be applied to the one or more antennas **1320**.

The processing system **1314** includes one or more processors **1304** coupled to a computer-readable medium/memory **1306**. The one or more processors **1304** are responsible for general processing, including the execution of software stored on the computer-readable medium/memory **1306**. The software, when executed by the one or more processors **1304**, causes the processing system **1314** to perform the various functions described supra for any particular apparatus. The computer-readable medium/memory **1306** may also be used for storing data that is manipulated by the one or more processors **1304** when executing software. The processing system **1314** further includes at least one of the reception component **1364**, the transmission

component 1370, the interference detection component 1376, the interference cancellation component 1378, and the configuration component 1382. The components may be software components running in the one or more processors 1304, resident/stored in the computer readable medium/memory 1306, one or more hardware components coupled to the one or more processors 1304, or some combination thereof. The processing system 1314 may be a component of the UE 350 and may include the memory 360 and/or at least one of the TX processor 368, the RX processor 356, and the communication processor 359.

In one configuration, the apparatus 1302 for wireless communication includes means for performing each of the operations of FIGS. 11(A) and 11(B). The aforementioned means may be one or more of the aforementioned components of the apparatus 1302 and/or the processing system 1314 of the apparatus 1302 configured to perform the functions recited by the aforementioned means.

As described supra, the processing system 1314 may include the TX Processor 368, the RX Processor 356, and the communication processor 359. As such, in one configuration, the aforementioned means may be the TX Processor 368, the RX Processor 356, and the communication processor 359 configured to perform the functions recited by the aforementioned means.

FIG. 14 is a diagram 1400 illustrating an example of a hardware implementation for an apparatus 1402 employing a processing system 1414. The apparatus 1402 may be a base station. The processing system 1414 may be implemented with a bus architecture, represented generally by a bus 1424. The bus 1424 may include any number of interconnecting buses and bridges depending on the specific application of the processing system 1414 and the overall design constraints. The bus 1424 links together various circuits including one or more processors and/or hardware components, represented by one or more processors 1404, a reception component 1464, a transmission component 1470, a cell configuration component 1476, a interference detection component 1478, and a computer-readable medium/memory 1406. The bus 1424 may also link various other circuits such as timing sources, peripherals, voltage regulators, and power management circuits, etc.

The processing system 1414 may be coupled to a transceiver 1410, which may be one or more of the transceivers 354. The transceiver 1410 is coupled to one or more antennas 1420, which may be the communication antennas 320.

The transceiver 1410 provides a means for communicating with various other apparatus over a transmission medium. The transceiver 1410 receives a signal from the one or more antennas 1420, extracts information from the received signal, and provides the extracted information to the processing system 1414, specifically the reception component 1464. In addition, the transceiver 1410 receives information from the processing system 1414, specifically the transmission component 1470, and based on the received information, generates a signal to be applied to the one or more antennas 1420.

The processing system 1414 includes one or more processors 1404 coupled to a computer-readable medium/memory 1406. The one or more processors 1404 are responsible for general processing, including the execution of software stored on the computer-readable medium/memory 1406. The software, when executed by the one or more processors 1404, causes the processing system 1414 to perform the various functions described supra for any particular apparatus. The computer-readable medium/memory

1406 may also be used for storing data that is manipulated by the one or more processors 1404 when executing software. The processing system 1414 further includes at least one of the reception component 1464, the transmission component 1470, the cell configuration component 1476, and the interference detection component 1478. The components may be software components running in the one or more processors 1404, resident/stored in the computer readable medium/memory 1406, one or more hardware components coupled to the one or more processors 1404, or some combination thereof. The processing system 1414 may be a component of the base station 310 and may include the memory 376 and/or at least one of the TX processor 316, the RX processor 370, and the controller/processor 375.

In one configuration, the apparatus 1402 for wireless communication includes means for performing each of the operations of FIG. 12. The aforementioned means may be one or more of the aforementioned components of the apparatus 1402 and/or the processing system 1414 of the apparatus 1402 configured to perform the functions recited by the aforementioned means.

As described supra, the processing system 1414 may include the TX Processor 316, the RX Processor 370, and the controller/processor 375. As such, in one configuration, the aforementioned means may be the TX Processor 316, the RX Processor 370, and the controller/processor 375 configured to perform the functions recited by the aforementioned means.

It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of exemplary approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean "one and only one" unless specifically so stated, but rather "one or more." The word "exemplary" is used herein to mean "serving as an example, instance, or illustration." Any aspect described herein as "exemplary" is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term "some" refers to one or more. Combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein

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by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

What is claimed is:

1. A method of wireless communication of a user equipment (UE), comprising:
 - receiving, from a first base station, a data transmission on a first cell using a first radio access technology (RAT);
 - determining a cell-specific reference signal (CRS) transmission using a second RAT on a second cell from a second base station;
 - determining that the CRS transmission on the second cell interferes with receiving the data transmission on the first cell at the UE, wherein determining that the CRS transmission interferes with receiving the data transmission at the UE includes:
 - receiving, from a network, a configuration for applying a given rate-matching to the data transmission to avoid interference from a transmission from a given base station using the second RAT, wherein determining that the first cell employs dynamic spectrum sharing (DSS) of the first RAT and the second RAT is based on receiving, from the network, a configuration for applying rate-matching to the data transmission to avoid interference from a CRS transmission on the first cell using the second RAT; and
 - applying an interference cancellation on the first cell to mitigate an interference from the CRS transmission on the second cell to the data transmission on the first cell.
2. The method of claim 1, further comprising:
 - determining that the rate-matching is not applied to the data transmission to avoid interference from the CRS transmission on the second cell to the data transmission on the first cell.
3. The method of claim 1, wherein the determination that the first cell employs DSS is further based on finding an identifier of the first cell in a record of DSS cells.

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4. The method of claim 1, wherein the determining that the CRS transmission interferes with receiving the data transmission at the UE includes:

receiving, from a network, an indication indicating that the CRS transmission on the second cell would interfere with receiving the data transmission on the first cell.

5. The method of claim 1, wherein the determining that the CRS transmission interferes with receiving the data transmission at the UE includes:

measuring a strength of the CRS transmission on the second cell.

6. The method of claim 1, further comprising:

determining a CRS configuration of the CRS transmission based on a cell configuration of the second cell, wherein the determining the CRS transmission is based on the CRS configuration.

7. The method of claim 1, further comprising:

receiving, from a network, a cell configuration of the second cell, wherein the cell configuration of the second cell includes at least one of: a physical cell ID (PCI), a CRS v-shift, a bandwidth, a CRS antenna port number, and a configuration of a Multicast Broadcast Single Frequency Network (MBSFN) subframe.

8. The method of claim 7, wherein the cell configuration of the second cell further includes at least one of a center frequency of the second cell and a timing offset.

9. The method of claim 1, further comprising:

receiving, from a network, a cell configuration of the second cell for applying the interference cancellation; and
monitoring the second cell based on the cell configuration of the second cell.

10. The method of claim 1, further comprising:

performing, while receiving the data transmission on the first cell, an inter-RAT measurement to detect the second cell; and

receiving a cell configuration transmitted on the second cell, wherein the application of the interference cancellation is based on the cell configuration.

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